

1. 3x3 Two-Dimensional Array

```
import java.util.Scanner;

public class Array3x3 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int arr[][] = new int[3][3];

        System.out.println("Enter values for 3x3 array:");

        for(int i=0;i<3;i++){
            for(int j=0;j<3;j++){
                arr[i][j] = sc.nextInt();
            }
        }

        System.out.println("Array Elements:");

        for(int i=0;i<3;i++){
            for(int j=0;j<3;j++){
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

2. Cinema Seat Booking System

```
import java.util.Scanner;

public class CinemaBooking {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int seats[][] = new int[3][3];
        int choice, row, col;

        do{
            System.out.println("\n1. Show Seats");
            System.out.println("2. Book Seat");
            System.out.println("3. Cancel Seat");
            System.out.println("4. Exit");

            System.out.print("Enter choice: ");
            choice = sc.nextInt();

            switch(choice){

                case 1:
                    System.out.println("Seating:");
                    for(int i=0;i<3;i++){
                        for(int j=0;j<3;j++){
                            System.out.print(seats[i][j]+" ");
                        }
                        System.out.println();
                    }
                    break;

                case 2:
                    System.out.print("Enter row and column: ");
```

```

        row = sc.nextInt();
        col = sc.nextInt();

        if(seats[row][col]==0){
            seats[row][col]=1;
            System.out.println("Seat Booked");
        }
        else{
            System.out.println("Seat already booked");
        }
        break;

    case 3:
        System.out.print("Enter row and column: ");
        row = sc.nextInt();
        col = sc.nextInt();

        if(seats[row][col]==1){
            seats[row][col]=0;
            System.out.println("Seat Cancelled");
        }
        else{
            System.out.println("Seat already empty");
        }
        break;

    case 4:
        System.out.println("Exiting...");
        break;
    }

    }while(choice!=4);
}

```

3. Cricket Score

```

import java.util.Scanner;

public class CricketScore {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int run[] = new int[5];
        int ball[] = new int[5];

        int total = 0;
        int highscore = 0;
        int player = 0;

        for(int i=0;i<5;i++){
            System.out.println("Player "+(i+1));
            System.out.print("Runs: ");
            run[i] = sc.nextInt();

            System.out.print("Balls: ");
            ball[i] = sc.nextInt();
        }

        for(int i=0;i<5;i++){
            total = total + run[i];

```

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        if(run[i] > highscore){
            highscore = run[i];
            player = i+1;
        }
    }

    System.out.println("\nScorecard:");

    for(int i=0;i<5;i++){
        double strikeRate = (run[i]*100.0)/ball[i];

        System.out.println("Player " +(i+1)+
            " Runs: "+run[i]+
            " Balls: "+ball[i]+
            " StrikeRate: "+strikeRate);
    }

    System.out.println("Total Runs = "+total);
    System.out.println("Highest Score = "+highscore);
    System.out.println("Player with Highest Score = "+player);
}
}

```

4. Count Digits, Letters & Symbols in String...

```

public class CountCharacters {

    public static void main(String[] args) {

        String s = "abc12f3#@";

        int digit = 0;
        int letter = 0;
        int symbol = 0;

        for(int i=0;i<s.length();i++){

            char c = s.charAt(i);

            if(Character.isDigit(c))
                digit++;

            else if(Character.isLetter(c))
                letter++;

            else
                symbol++;
        }

        System.out.println("Digits: "+digit);
        System.out.println("Letters: "+letter);
        System.out.println("Symbols: "+symbol);
    }
}

```

5. Password Validation

```

public class PasswordCheck {

    public static void main(String[] args) {

        String password = "Abc@123";
    }
}

```

```

boolean hasUpper=false;
boolean hasLower=false;
boolean hasDigit=false;
boolean hasSymbol=false;

if(password.length()>=6 && password.length()<=16){
    for(int i=0;i<password.length();i++){
        char c = password.charAt(i);

        if(Character.isUpperCase(c))
            hasUpper=true;

        else if(Character.isLowerCase(c))
            hasLower=true;

        else if(Character.isDigit(c))
            hasDigit=true;

        else if(c=='@' || c=='#' || c=='$')
            hasSymbol=true;
    }

    if(hasUpper && hasLower && hasDigit && hasSymbol)
        System.out.println("Valid Password");

    else
        System.out.println("Invalid Password");
}
else{
    System.out.println("Password length must be 6 to 16");
}
}

```